

SAFETY DATA SHEET

TORAC® Insecticide

Section 1. Identification

GHS product identifier	: TORAC® Insecticide
Other means of identification	: Active ingredient: Common name: Tolfenpyrad Chemical name: 1H-Pyrazole-5-carboxamide, 4-chloro-3-ethyl-1-methyl-N-[[4-(4-methylphenoxy)phenyl]methyl]-
Product code	: EPA Registration Number: 71711-31
Product use	: Insecticide. Emulsifiable concentrate (EC)
Supplier's details	: Nichino America Inc 4550 Linden Hill Road, Suite 501 Wilmington, DE 19808 United States Telephone number: 302-636-9001
e-mail address of person responsible for this SDS	: Not available.
Emergency telephone number (with hours of operation)	: In case of fire or spill: (800) 424-9300 (24 hours per day) For international shipments: (703) 527-3887 (24 hours per day) For emergency health and safety inquiries: (800) 348-5832 (24 hours per day)

Section 2. Hazards identification

OSHA/HCS status	: This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
Classification of the substance or mixture	: ACUTE TOXICITY (oral) - Category 3 ACUTE TOXICITY (inhalation) - Category 3 SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A CARCINOGENICITY - Category 2 TOXIC TO REPRODUCTION - Category 1B SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 1 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 2 ASPIRATION HAZARD - Category 1

GHS label elements

Hazard pictograms



Signal word

: Danger

Hazard statements

: Toxic if swallowed or if inhaled.
May be fatal if swallowed and enters airways.
Causes skin irritation.
Causes serious eye irritation.
Suspected of causing cancer.
May damage fertility or the unborn child.
Causes damage to organs through prolonged or repeated exposure. (liver, pancreas, reproductive organs) (oral)
May cause damage to organs through prolonged or repeated exposure. (heart) (oral)

Precautionary statements

Section 2. Hazards identification

Prevention	: Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear protective gloves, protective clothing and eye or face protection. Use only outdoors or in a well-ventilated area. Do not breathe dust or mist. Do not eat, drink or smoke when using this product. Wash hands thoroughly after handling.
Response	: IF exposed or concerned: Get medical advice or attention. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or doctor. IF SWALLOWED: Immediately call a POISON CENTER or doctor. Rinse mouth. Do NOT induce vomiting. Take off contaminated clothing and wash it before reuse. IF ON SKIN: Wash with plenty of water. If skin irritation occurs: Get medical advice or attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice or attention.
Storage	: Store locked up. Store in a well-ventilated place. Keep container tightly closed.
Disposal	: Dispose of contents and container in accordance with all local, regional, national and international regulations.
Hazards not otherwise classified	: None known.
Hazards identified when used	: No known significant effects or critical hazards.

Section 3. Composition/information on ingredients

Substance/mixture	: Mixture
Other means of identification	: Active ingredient: Common name: Tolfenpyrad Chemical name: 1H-Pyrazole-5-carboxamide, 4-chloro-3-ethyl-1-methyl-N-[[4-(4-methylphenoxy)phenyl]methyl]-
Product code	: EPA Registration Number: 71711-31

Ingredient name	Synonyms	%	Identifiers
 Tolfenpyrad	4-Chloro-3-ethyl-1-methyl-N-[[4-(4-methylphenoxy)phenyl]methyl]-1H-pyrazole-5-carboxamide; N-[4-(p-Tolyloxy)benzyl]-1-methyl-3-ethyl-4-chloro-1h-pyrazole-5-carboxamide; 4-Chloro-3-ethyl-1-methyl-N-[4-(4-tolyloxy)benzyl]-1H-pyrazole-5-carboxamide;	15	CAS: 129558-76-5
N-methyl-2-pyrrolidone	1-methyl-2-pyrrolidone; 2-Pyrrolidinone, 1-methyl-; n-Methylpyrrolidone; NMP; N-Methyl-2-pyrrolidinone; 1-methylpyrrolidin-2-one; dielectric coating, buffering mechanical stress, consisting of a radically photopatternable polyamide-precursor with unsaturated carbon in the side-chains which is convertible into a polyimide, in form of a solution from N-methyl-2-pyrrolidone (CAS RN 872-50-4) or N-ethyl-2-pyrrolidon (CAS RN 2687-91-4) with a polymer content by weight of more than 10 % or more; 1-methyl-2-pyrrolidinone; M-pyrol; Methyl-2-Pyrrolidone, n-; n-methyl-2-pyrrolidine	≥10 - ≤30	CAS: 872-50-4

Section 3. Composition/information on ingredients

Solvent naphtha (petroleum), heavy arom.	Kerosine - unspecified; Solvent naphtha, petroleum, heavy aromatic; (Polyethyl)benzenes; Solvent naphtha, petroleum, heavy arom. - ultra low naphthalene; Heavy aromatic solvent naphtha; Solvent naphtha; Solvent naphtha (petroleum), heavy aromatic; Heavy solvent naphtha; Solvent naphtha (petroleum), heavy arom; AROMATIC PETROLEUM DISTILLATE; Solvent Naphtha (petroleum)	≥45 - ≤70	CAS: 64742-94-5
2-methylnaphthalene	Naphthalene, 2-methyl-; .beta.-Methylnaphthalene; naphthylene, 2-methyl-; 2-methyl naphthalene [PAH, POM]; beta-Methylnaphthalene; 2-methyl-naphthalene; Mono and dimethylnaphthalene; β-Methyl naphthalene	≥10 - ≤30	CAS: 91-57-6
naphthalene	White tar; Tar camphor; Naphthalin; naphthalene, pure; naphthalene, crude; MOTH FLAKES; Naphthalene (8CA & 9CA); naphthalene [PAH, POM]; NAPHTHALENE, REFINED; NAPHTHALENE, MOLTEN; naphthalene	≥5 - ≤10	CAS: 91-20-3
1-methylnaphthalene	Naphthalene, 1-methyl-; .alpha.-Methylnaphthalene; alpha-Methylnaphthalene; alpha-Methylnaftalen; α-Methylnaphthalene; Mono and dimethylnaphthalene; METHYL NAPHTHALENE; 1-methylnaphthalene	≥3 - ≤7	CAS: 90-12-0

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified and hence require reporting in this section.

Occupational exposure limits, if available, are listed in Section 8.

Section 4. First aid measures

Description of necessary first aid measures

- | | |
|--------------------|---|
| Eye contact | : Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Continue to rinse for at least 10 minutes. Get medical attention. |
| Inhalation | : Remove victim to fresh air and keep at rest in a position comfortable for breathing. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Get medical attention. If necessary, call a poison center or physician. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband. In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed |

Section 4. First aid measures

- person may need to be kept under medical surveillance for 48 hours.
- Skin contact** : Flush contaminated skin with plenty of water. Remove contaminated clothing and shoes. Wash contaminated clothing thoroughly with water before removing it, or wear gloves. Continue to rinse for at least 10 minutes. Get medical attention. Wash clothing before reuse. Clean shoes thoroughly before reuse.
- Ingestion** : Get medical attention immediately. Call a poison center or physician. Wash out mouth with water. Remove dentures if any. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Stop if the exposed person feels sick as vomiting may be dangerous. Aspiration hazard if swallowed. Can enter lungs and cause damage. Do not induce vomiting. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband.

Most important symptoms/effects, acute and delayed

Potential acute health effects

- Eye contact** : Causes serious eye irritation.
- Inhalation** : Toxic if inhaled.
- Skin contact** : Causes skin irritation.
- Ingestion** : Toxic if swallowed. May be fatal if swallowed and enters airways.

Over-exposure signs/symptoms

- Eye contact** : Adverse symptoms may include the following:
pain or irritation
watering
redness
- Inhalation** : Adverse health effects could include the following:
respiratory tract irritation
coughing
- Skin contact** : Adverse symptoms may include the following:
irritation
redness
- Ingestion** : Adverse symptoms may include the following:
nausea or vomiting
reduced fetal weight
increase in fetal deaths
skeletal malformations

Indication of immediate medical attention and special treatment needed, if necessary

- Notes to physician** : In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.
- Specific treatments** : No specific treatment.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

See toxicological information (Section 11)

Section 5. Fire-fighting measures

Extinguishing media

- Suitable extinguishing media** : Use dry chemical, CO₂, water spray (fog) or foam. Use an extinguishing agent suitable for the surrounding fire.
- Unsuitable extinguishing media** : Do not use water jet.

Specific hazards arising from the chemical : In a fire or if heated, a pressure increase will occur and the container may burst.

Hazardous thermal decomposition products : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
nitrogen oxides
halogenated compounds

Special protective actions for fire-fighters : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.

Special protective equipment for fire-fighters : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Section 6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

For non-emergency personnel : No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Do not breathe vapor or mist. Provide adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Put on appropriate personal protective equipment.

For emergency responders : If specialized clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".

Environmental precautions : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

Methods and materials for containment and cleaning up

Small spill : Stop leak if without risk. Move containers from spill area. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.

Large spill : Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Dispose of via a licensed waste disposal contractor. Contaminated absorbent material may pose the same hazard as the spilled product. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations.

Section 7. Handling and storage

Precautions for safe handling

Protective measures : Put on appropriate personal protective equipment (see Section 8). Avoid exposure - obtain special instructions before use. Avoid exposure during pregnancy. Do not handle until all safety precautions have been read and understood. Do not get in eyes or on skin or clothing. Do not breathe vapor or mist. Do not swallow. Use only with adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Empty containers retain product residue and can be

Section 7. Handling and storage

Advice on general occupational hygiene

hazardous. Do not reuse container.

- : Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

Conditions for safe storage, including any incompatibilities

- : Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Store locked up. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination. See Section 10 for incompatible materials before handling or use.

Section 8. Exposure controls/personal protection

Control parameters

Occupational exposure limits

Ingredient name	Exposure limits
Solvent naphtha (petroleum), heavy arom.	None.
Tolfenpyrad	None.
2-methylnaphthalene	ACGIH TLV (United States, 1/2025) A4. Absorbed through skin. TWA 8 hours: 0.5 ppm.
N-methyl-2-pyrrolidone	OARS WEEL (United States, 6/2025) Absorbed through skin. TWA 8 hours: 15 ppm. STEL 15 minutes: 120 mg/m ³ . STEL 15 minutes: 30 ppm. TWA 8 hours: 60 mg/m ³ . CAL OSHA PEL (United States, 1/2025) Absorbed through skin. TWA 8 hours: 4 mg/m ³ . TWA 8 hours: 1 ppm.
naphthalene	NIOSH REL (United States, 10/2020) TWA 10 hours: 10 ppm. TWA 10 hours: 50 mg/m ³ . STEL 15 minutes: 15 ppm. STEL 15 minutes: 75 mg/m ³ . CAL OSHA PEL (United States, 1/2025) Absorbed through skin. TWA 8 hours: 0.5 mg/m ³ . TWA 8 hours: 0.1 ppm. OSHA PEL (United States, 5/2018) TWA 8 hours: 10 ppm. TWA 8 hours: 50 mg/m ³ . OSHA PEL 1989 (United States, 3/1989) TWA 8 hours: 10 ppm. TWA 8 hours: 50 mg/m ³ . STEL 15 minutes: 15 ppm. STEL 15 minutes: 75 mg/m ³ . ACGIH TLV (United States, 1/2025) A3. Absorbed through skin. TWA 8 hours: 10 ppm. TWA 8 hours: 52 mg/m ³ .

Section 8. Exposure controls/personal protection

1-methylnaphthalene

ACGIH TLV (United States, 1/2025) A4.
Absorbed through skin.
TWA 8 hours: 0.5 ppm.

Biological exposure indices

Ingredient name	Exposure indices
2-methylnaphthalene	ACGIH BEI (United States, 1/2025) [polycyclic aromatic hydrocarbons] BEI: 2.5 µg/l, 1-hydroxypyrene [in urine]. Sampling time: end of shift at end of workweek. BEI: Nonquantitative: Biological monitoring should be considered for this compound based on the review; however, a specific BEI® could not be determined due to insufficient data., 3-hydroxybenzo(a)pyrene [in urine]. Sampling time: end of shift at end of workweek.
N-methyl-2-pyrrolidone	ACGIH BEI (United States, 1/2025) BEI: 100 mg/l, 5-hydroxy-N-methyl-2-pyrrolidone [in urine]. Sampling time: end of shift.
naphthalene	ACGIH BEI (United States, 1/2025) BEI: Nonquantitative: Biological monitoring should be considered for this compound based on the review; however, a specific BEI® could not be determined due to insufficient data., 1-naphthol + 2-naphthol [(sample not specified)]. Sampling time: end of shift.
1-methylnaphthalene	ACGIH BEI (United States, 1/2025) [polycyclic aromatic hydrocarbons] BEI: 2.5 µg/l, 1-hydroxypyrene [in urine]. Sampling time: end of shift at end of workweek. BEI: Nonquantitative: Biological monitoring should be considered for this compound based on the review; however, a specific BEI® could not be determined due to insufficient data., 3-hydroxybenzo(a)pyrene [in urine]. Sampling time: end of shift at end of workweek.

Appropriate engineering controls

: Use only with adequate ventilation. Use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits.

Environmental exposure controls

: Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

Individual protection measures

Hygiene measures

: Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Section 8. Exposure controls/personal protection

- Eye/face protection** : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles.
- Skin protection**
- Hand protection** : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary. Considering the parameters specified by the glove manufacturer, check during use that the gloves are still retaining their protective properties. It should be noted that the time to breakthrough for any glove material may be different for different glove manufacturers. In the case of mixtures, consisting of several substances, the protection time of the gloves cannot be accurately estimated.
- Body protection** : Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.
- Other skin protection** : Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.
- Respiratory protection** : Based on the hazard and potential for exposure, select a respirator that meets the appropriate standard or certification. Respirators must be used according to a respiratory protection program to ensure proper fitting, training, and other important aspects of use.
Recommended: Ensure an MSHA/NIOSH-approved respirator or equivalent is used.

Section 9. Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

Appearance

- Physical state** : Liquid.
- Color** : Pale yellow.
- Odor** : Hydrocarbon.
- Odor threshold** : Not available.
- pH** : 6.1 [Conc. (% w/w): 1%] [emulsion]
- Melting point/freezing point** : Not available.
- Boiling point or initial boiling point and boiling range** : Not available.
- Flash point** : 96°C (204.8°F)
- Evaporation rate** : Not available.
- Flammability** : Not available.
- Lower and upper explosion limit/flammability limit** : Not available.
- Vapor pressure** : Not available.
- Relative vapor density** : Not available.
- Relative density** : Not available.
- Density** : 1.03 g/cm³ [20°C (68°F)]
- Solubility in water** : Not available.
- Partition coefficient: n-octanol/water** : Not applicable.
- Auto-ignition temperature** : Not available.
- Decomposition temperature** : Not available.
- Viscosity** : Dynamic (room temperature): Not available.
Kinematic (room temperature): 12.9 mm²/s (12.9 cSt) [20°C]
Kinematic (40°C (104°F)): 6.52 mm²/s (6.52 cSt)
- Explosive properties** : Not available.

Section 9. Physical and chemical properties

Oxidizing properties : Not available.

Particle characteristics

Median particle size : Not applicable.

Section 10. Stability and reactivity

Reactivity : No specific test data related to reactivity available for this product or its ingredients.

Chemical stability : Stable under recommended storage and handling conditions (see Section 7).

Possibility of hazardous reactions : Under normal conditions of storage and use, hazardous polymerization will not occur.

Conditions to avoid : Keep away from heat, sparks and flame.

Incompatible materials : No specific data.

Hazardous decomposition products : Hazardous decomposition products (In case of fire):
carbon dioxide
carbon monoxide
nitrogen oxides

Section 11. Toxicological information

Information on toxicological effects

Acute toxicity

Product/ingredient name	Result and Species	Dose [Exposure]	Remarks
TORAC® Insecticide	Oral - Rat - Female - LD50	83 mg/kg	-
	Oral - Rat - Male - LD50	102 mg/kg	-
	Dermal - Rat - LD50	>2000 mg/kg	-
	Inhalation - Rat - LC50 Dusts and mists	0.542 mg/l [4 hours]	-
	Solvent naphtha (petroleum), heavy arom.	Oral - Rat - LD50 [OECD 401]	>5000 mg/kg
	Dermal - Rabbit - LD50 [OECD 402]	>2000 mg/kg	-
Tolfenpyrad	Inhalation - Rat - LC50 Dusts and mists [OECD 403]	>4778 mg/m³ [4 hours]	-
	Dermal - Rat - Male - LD50	>2000 mg/kg	no mortality
	Oral - Rat - Male - LD50	386 mg/kg	-
	Oral - Rat - Female - LD50	150 mg/kg	-
	Dermal - Rat - Female - LD50	>3000 mg/kg	no mortality
	Inhalation - Rat - Male - LC50 Dusts and mists	2.21 mg/l [4 hours]	-
	Inhalation - Rat - Female - LC50 Dusts and mists	1.5 mg/l [4 hours]	-

Conclusion/Summary : Toxic if swallowed or if inhaled.

Section 11. Toxicological information

Irritation/Corrosion

Product/ingredient name	Result and Species	Exposure	Remarks
TORAC® Insecticide Solvent naphtha (petroleum), heavy arom. Tolfenpyrad	Eyes - Rabbit - Irritant	-	-
	Skin - Rabbit - Irritant	-	-
	Skin - Unknown species - Mild irritant - [OECD 404]	-	(similar material)
	Eyes - Unknown species - Mild irritant - [OECD 405]	-	(similar material)
	Skin - Rabbit - Slightly irritating to the skin.	-	-
	Eyes - Rabbit - Slightly irritating to the eyes.	-	-

Conclusion/Summary

Skin : Causes skin irritation.
Eyes : Causes serious eye irritation.
Respiratory : Not available.

Respiratory or skin sensitization

Product/ingredient name	Route of exposure and Species	Result	Remarks
TORAC® Insecticide	skin - Guinea pig	Not sensitizing	-
Solvent naphtha (petroleum), heavy arom.	skin - Unknown species [OECD 406]	Not sensitizing	(similar material)
Tolfenpyrad	skin - Guinea pig	Not sensitizing	-

Conclusion/Summary

Skin : Based on available data, the classification criteria are not met.
Respiratory : Not available.

Mutagenicity

Product/ingredient name	Result	Experiment	Remarks
Solvent naphtha (petroleum), heavy arom.	Negative [OECD 473, 474, 475, 476, 479]	In vitro - Bacteria	(similar material)
Tolfenpyrad	Negative [Bacterial Reverse Mutation Test]	Bacteria	-
	Negative [In vitro Mammalian Chromosomal Aberration Test]	In vitro - Mammalian-Animal	-
	Negative [Micronucleus-test]	Mammalian-Animal	-

Conclusion/Summary : Based on available data, the classification criteria are not met.

Carcinogenicity

Product/ingredient name	Result	Species and Route of exposure	Dose [Exposure]	Remarks
Tolfenpyrad	Negative	Rat - Oral -	2 years	-
	Negative	Mouse - Oral -	18 months	-

Section 11. Toxicological information

Conclusion/Summary : Suspected of causing cancer.

Classification

Product/ingredient name	OSHA	IARC	NTP
naphthalene	-	2B	Reasonably anticipated to be a human carcinogen.

Reproductive toxicity

Product/ingredient name	Result	Species and Route of exposure	Dose [Exposure]	Remarks
	<u>Maternal toxicity</u> : Negative <u>Fertility effects</u> : Negative <u>Developmental</u> : Negative [OECD 414, 416]	Unknown species - Route of exposure unreported	-	(similar material)
Tolfenpyrad	<u>Developmental</u> : Negative	Rat - Unreported	-	-
	<u>Developmental</u> : Negative	Rabbit - Unreported	-	-
N-methyl-2-pyrrolidone	<u>Maternal toxicity</u> : Positive <u>Fertility effects</u> : Positive <u>Developmental</u> : Positive	Rat - Unreported	-	-

Conclusion/Summary : May damage fertility or the unborn child.

Teratogenicity

Conclusion/Summary : Not available.

Specific target organ toxicity (single exposure)

Name	Category	Route of exposure	Target organs
N-methyl-2-pyrrolidone	Category 3	-	Respiratory tract irritation

Specific target organ toxicity (repeated exposure)

Name	Category	Route of exposure	Target organs
	Category 1	oral	liver, pancreas, reproductive organs
	Category 2	oral	heart
Tolfenpyrad	Category 1	oral	liver, pancreas, reproductive organs
	Category 2	oral	heart

Aspiration hazard

Name	Result
	ASPIRATION HAZARD - Category 1

Information on the likely routes of exposure : Routes of entry anticipated: Oral, Dermal, Inhalation, Eyes.

Potential acute health effects

Eye contact : Causes serious eye irritation.

Inhalation : Toxic if inhaled.

Skin contact : Causes skin irritation.

Ingestion : Toxic if swallowed. May be fatal if swallowed and enters airways.

Section 11. Toxicological information

Symptoms related to the physical, chemical and toxicological characteristics

- Eye contact** : Adverse symptoms may include the following:
pain or irritation
watering
redness
- Inhalation** : Adverse health effects could include the following:
respiratory tract irritation
coughing
- Skin contact** : ☒ Adverse symptoms may include the following:
irritation
redness
- Ingestion** : Adverse symptoms may include the following:
nausea or vomiting
reduced fetal weight
increase in fetal deaths
skeletal malformations

Delayed and immediate effects and also chronic effects from short and long term exposure

Short term exposure

- Potential immediate effects** : Not available.
- Potential delayed effects** : Not available.

Long term exposure

- Potential immediate effects** : Not available.
- Potential delayed effects** : Not available.

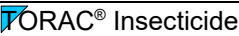
Potential chronic health effects

Product/ingredient name	Result and Species	Dose [Exposure]	Remarks
Tolfenpyrad	Oral - Chronic - Rat - -	4.78 to 9.33 mg/kg/day [90 days]	-
	Oral - Chronic - Mouse - -	15.9 to 46.2 mg/kg/day [90 days]	-

- Conclusion/Summary** : Causes damage to organs through prolonged or repeated exposure if inhaled. (liver, pancreas, reproductive organs) (Oral)
- General** : Causes damage to organs through prolonged or repeated exposure if swallowed.
- Carcinogenicity** : Suspected of causing cancer. Risk of cancer depends on duration and level of exposure.
- Mutagenicity** : No known significant effects or critical hazards.
- Reproductive toxicity** : May damage fertility or the unborn child.

Numerical measures of toxicity

Acute toxicity estimates

Product/ingredient name	Oral (mg/kg)	Dermal (mg/kg)	Inhalation (gases) (ppm)	Inhalation (vapors) (mg/l)	Inhalation (dusts and mists) (mg/l)
 TORAC® Insecticide	83	2500	N/A	N/A	0.542
Solvent naphtha (petroleum), heavy arom.	N/A	2500	N/A	N/A	N/A
Tolfenpyrad	150	2500	N/A	N/A	1.5
naphthalene	500	N/A	N/A	N/A	N/A

Section 11. Toxicological information

Section 12. Ecological information

Toxicity

Product/ingredient name	Result [Exposure]	Species	Remarks
Tolfenpyrad	Acute - LC50 0.0029 mg/l [96 hours]	Fish - <i>Cyprinus carpio</i>	-
	Acute - EC50 0.001 mg/l [48 hours]	Daphnia - <i>Daphnia magna</i>	-
	Acute - ErC50 >0.76 mg/l [72 hours]	Algae - <i>Raphidocelis subcapitata</i>	-
	Acute - LC50 0.000163 mg/l [96 hours]	Fish - <i>Oncorhynchus mykiss</i>	-
	Chronic - NOEC 0.000244 mg/l [21 days]	Daphnia - <i>Daphnia magna</i>	-
	Chronic - NOEC 0.000188 mg/l [33 days]	Fish - <i>Pimephales promelas</i>	-
	Chronic - NOEC >0.76 mg/l [72 hours]	Algae - <i>Raphidocelis subcapitata</i>	-

Conclusion/Summary : Very toxic to aquatic life with long lasting effects.

Persistence and degradability

Product/ingredient name	Test	Result	Remarks
Solvent naphtha (petroleum), heavy arom.	-	60.74% [28 days] - Readily	-

Conclusion/Summary : There are no data available on the mixture itself.

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
Solvent naphtha (petroleum), heavy arom.	-	-	Readily

Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
Tolfenpyrad	-	2905 to 3007	High

Mobility in soil

Soil/Water partition coefficient : Not available.

Mobility : Not available.

Other adverse effects : No known significant effects or critical hazards.

Section 13. Disposal considerations

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

PESTICIDE DISPOSAL: Wastes resulting from the use of this product must be disposed of on site or at an approved waste disposal facility.

RCRA Toxic hazardous waste "U" List

Ingredient	CAS #	Status	Reference number
Naphthalene	91-20-3	Listed	U165

Section 14. Transport information

	DOT Classification	TDG Classification	Mexico Classification	ADR/RID	IMDG	IATA
UN number	UN2902	UN2902	UN2902	UN2902	UN2902	UN2902
UN proper shipping name	Pesticides, liquid, toxic, n. o.s. (Tolfenpyrad)	PESTICIDE, LIQUID, TOXIC, N.O.S. (Tolfenpyrad)	PLAGUICIDA, LÍQUIDO, TÓXICO, N.E. P. (Tolfenpyrad)	PESTICIDE, LIQUID, TOXIC, N.O.S. (Tolfenpyrad)	PESTICIDE, LIQUID, TOXIC, N.O.S. (Tolfenpyrad)	Pesticide, liquid, toxic, n. o.s. (Tolfenpyrad)
Transport hazard class(es)	6.1	6.1	6.1	6.1	6.1	6.1
Label				 	 	
Packing group	III	III	III	III	III	III
Environmental hazards	No.	No.	No.	Yes.	Marine Pollutant: Yes	Yes. The environmentally hazardous substance mark is not required.

Additional information

Section 14. Transport information

DOT Classification	: Reportable quantity 1324.2 lbs / 601.2 kg [154.19 gal / 583.69 L]. Package sizes shipped in quantities less than the product reportable quantity are not subject to the RQ (reportable quantity) transportation requirements. Limited quantity Yes. Packaging instruction Exceptions: 153. Non-bulk: 203. Bulk: 241. Quantity limitation Passenger aircraft/rail: 60 L. Cargo aircraft: 220 L. Special provisions IB3, T7, TP2, TP28
TDG Classification	: Product classified as per the following sections of the Transportation of Dangerous Goods Regulations: 2.26-2.36 (Class 6). Explosive Limit and Limited Quantity Index 5 Passenger Carrying Road or Rail Index 60 Special provisions 16
Mexico Classification	: Special provisions 61, 223, 274
ADR/RID	: The environmentally hazardous substance mark is not required when transported in sizes of ≤5 L or ≤5 kg. Hazard identification number 60 Limited quantity 5 L Special provisions 61, 274, 648 Tunnel code (E)
IMDG	: The marine pollutant mark is not required when transported in sizes of ≤5 L or ≤5 kg. Emergency schedules F-A, S-A Special provisions 61, 223, 274
IATA	: The environmentally hazardous substance mark may appear if required by other transportation regulations. Quantity limitation Passenger and Cargo Aircraft: 60 L. Packaging instructions: 655. Cargo Aircraft Only: 220 L. Packaging instructions: 663. Limited Quantities - Passenger Aircraft: 2 L. Packaging instructions: Y642. Special provisions A3, A4
Special precautions for user	: Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Transport in bulk according to IMO instruments : Not intended.

Section 15. Regulatory information

U.S. Federal regulations

TSCA 6 proposed risk management: N-methyl-2-pyrrolidone

TSCA 8(a) PAIR: naphthalene

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

TSCA 8(d) H and S data reporting: naphthalene

Clean Water Act (CWA) 307: 2-methylnaphthalene; naphthalene; 1-methylnaphthalene

This chemical is a pesticide product registered by the United States Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets (SDS), and for workplace labels of non-pesticide chemicals. The hazard information required on the pesticide label is reproduced below. The pesticide label also includes other important information, including directions for use.

EPA Registration Number: 71711-31

WARNING

May be fatal if swallowed.

Causes skin irritation.

Causes substantial but temporary eye injury.

Do not get in eyes, on skin or on clothing.

Section 15. Regulatory information

Harmful if absorbed through skin.

Harmful if inhaled.

Avoid breathing the spray mist.

Remove and wash contaminated clothing before reuse.

Wash thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco, or using the toilet

TSCA 12(b) - Chemical export notification

Name	One time notification		Annual notification		
	4	5	5(f)	6	7
n-methylpyrrolidone	Not listed	Not listed	Not listed	Listed	Not listed

Clean Air Act Section 112 : Listed

(b) Hazardous Air Pollutants (HAPs)

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304

Composition/information on ingredients

No products were found.

SARA 304 RQ : Not applicable.

SARA 311/312

Classification : ACUTE TOXICITY (oral) - Category 3
 ACUTE TOXICITY (inhalation) - Category 3
 SKIN IRRITATION - Category 2
 EYE IRRITATION - Category 2A
 CARCINOGENICITY - Category 2
 TOXIC TO REPRODUCTION - Category 1B
 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 1
 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 2
 ASPIRATION HAZARD - Category 1

Composition/information on ingredients

Name	%	Classification
Tolfenpyrad	15	ACUTE TOXICITY (oral) - Category 3 ACUTE TOXICITY (inhalation) - Category 4 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 1 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 2
Solvent naphtha (petroleum), heavy arom.	≥45 - ≤70	CARCINOGENICITY - Category 2 ASPIRATION HAZARD - Category 1
N-methyl-2-pyrrolidone	≥10 - ≤30	FLAMMABLE LIQUIDS - Category 4 SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A TOXIC TO REPRODUCTION - Category 1B SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
naphthalene	≥5 - ≤10	ACUTE TOXICITY (oral) - Category 4

Section 15. Regulatory information

CARCINOGENICITY - Category 2

SARA 313

	Product name	CAS number	%
Form R - Reporting requirements	N-methyl-2-pyrrolidone naphthalene	872-50-4 91-20-3	≥10 - ≤30 ≥5 - ≤10
Supplier notification	N-methyl-2-pyrrolidone naphthalene	872-50-4 91-20-3	≥10 - ≤30 ≥5 - ≤10

SARA 313 notifications must not be detached from the SDS and any copying and redistribution of the SDS shall include copying and redistribution of the notice attached to copies of the SDS subsequently redistributed.

State regulations

- Massachusetts** : The following components are listed: 1-METHYL-2-PYRROLIDONE; NAPHTHALENE; 1-METHYLNAPHTHALENE; 2-ETHYLHEXANOL
- New York** : The following components are listed: Naphthalene
- New Jersey** : The following components are listed: 2-METHYL NAPHTHALENE; 1-METHYL-2-PYRROLIDONE; NAPHTHALENE; 1-METHYL NAPHTHALENE
- Pennsylvania** : The following components are listed: 2-PYRROLIDINONE, 1-METHYL-; NAPHTHALENE; NAPHTHALENE, 1-METHYL-; 1-HEXANOL, 2-ETHYL-

California Prop. 65

WARNING: This product can expose you to chemicals including Naphthalene, which is known to the State of California to cause cancer, and N-methylpyrrolidone, which is known to the State of California to cause birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

Ingredient name	No significant risk level	Maximum acceptable dosage level
N-methylpyrrolidone Naphthalene	- Yes.	Yes. -

EPA PFAS Compilation from Comptox

Not listed.

TSCA 8(a)7 - One-time Reporting PFAS

Not listed.

International regulations

Chemical Weapon Convention List Schedules I, II & III Chemicals

Not listed.

Montreal Protocol

Not listed.

Stockholm Convention on Persistent Organic Pollutants

Not listed.

Rotterdam Convention on Prior Informed Consent (PIC)

Not listed.

UNECE Aarhus Protocol on POPs and Heavy Metals

Not listed.

OECD Comprehensive Global PFAS Database

Not listed.

Inventory list

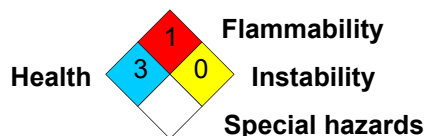
- Eurasian Economic Union** : **Russian Federation inventory:** All components are listed or exempted.
- Taiwan** : All components are listed or exempted.

Section 15. Regulatory information

United States : All components are active or exempted. This product is a registered pesticide.
Viet Nam : All components are listed or exempted.

Section 16. Other information

National Fire Protection Association (U.S.A.)



Procedure used to derive the classification

Classification	Justification
ACUTE TOXICITY (oral) - Category 3	On basis of test data
ACUTE TOXICITY (inhalation) - Category 3	On basis of test data
SKIN IRRITATION - Category 2	On basis of test data
EYE IRRITATION - Category 2A	On basis of test data
CARCINOGENICITY - Category 2	Calculation method
TOXIC TO REPRODUCTION - Category 1B	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 1	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 2	Calculation method
ASPIRATION HAZARD - Category 1	On basis of test data

History

Date of printing : 01/30/2026
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Version : 2
Key to abbreviations : ADR = Agreement concerning the International Carriage of Dangerous Goods by Road
ATE = Acute Toxicity Estimate
BCF = Bioconcentration Factor
DOT = Department of Transportation
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
IATA = International Air Transport Association
IBC = Intermediate Bulk Container
IMDG = International Maritime Dangerous Goods
LogPow = logarithm of the octanol/water partition coefficient
MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution)
N/A = Not available
RID = The Regulations concerning the International Carriage of Dangerous Goods by Rail
SGG = Segregation Group
TDG = Transportation of Dangerous Goods
UN = United Nations

References : Not available.

Indicates information that has changed from previously issued version.

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